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Studierichting: Informatica
Oefeningenreeks 2E nr. 4.21

$$
\begin{aligned}
&\lim_{x\to\infty}\left(\frac{x^2-2}{x^2-1}\right)^{2x^2+x+1}\\
&=\lim_{x\to\infty}\left(\frac{x^2-1-1}{x^2-1}\right)^{2x^2+x+1}\\
&=\lim_{x\to\infty}\left(1-\frac{1}{x^2-1}\right)^{2x^2+x+1}\\
&=\lim_{x\to\infty}\left(\left(1+\frac{-1}{x^2-1}\right)^{\frac{x^2-1}{-1}\cdot\frac{-1}{x^2-1}}\right)^{2x^2+x+1}\\
&=e^{\lim_{x\to\infty}\frac{-(2x^2+x+1)}{x^2-1}}\\
&=e^{\lim_{x\to\infty}\frac{-2x^2}{x^2}}\\
&=e^{-2}\\
&=\frac{1}{e^2}
\end{aligned}
$$

References